



Structure Of Bacterial Cells

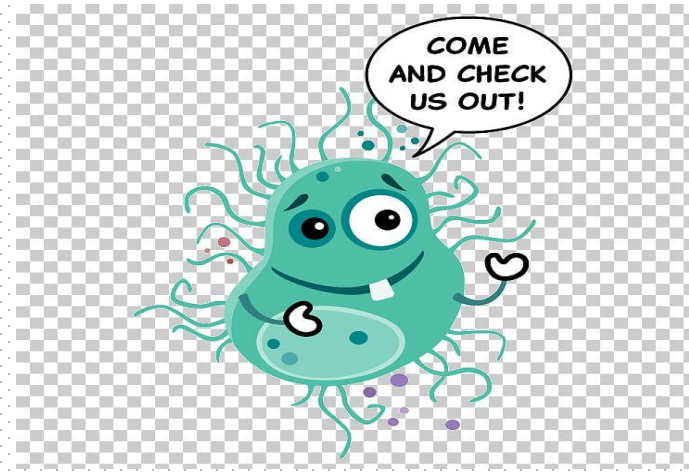
(SGL 2)

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Introduction



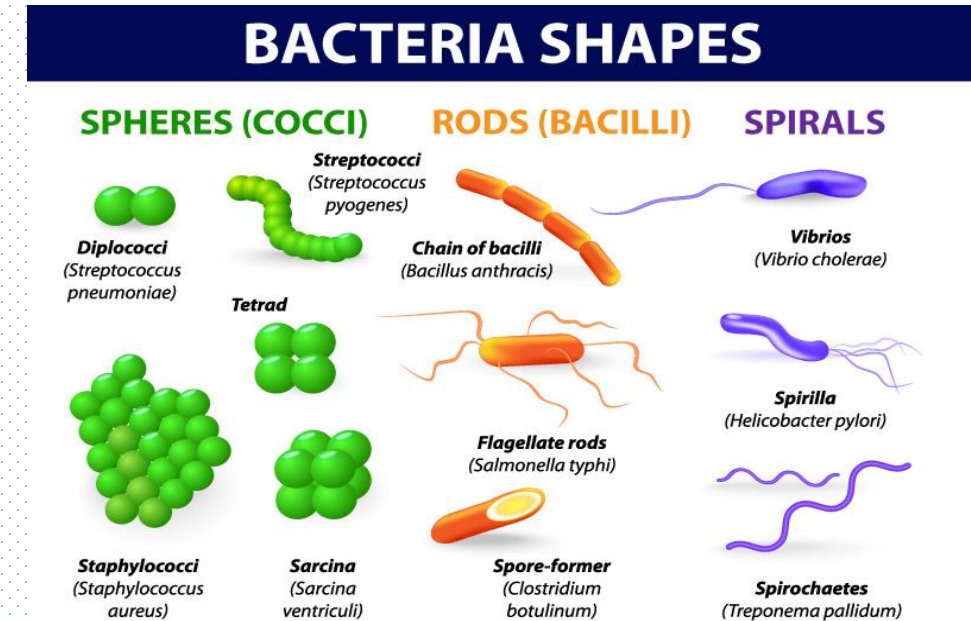
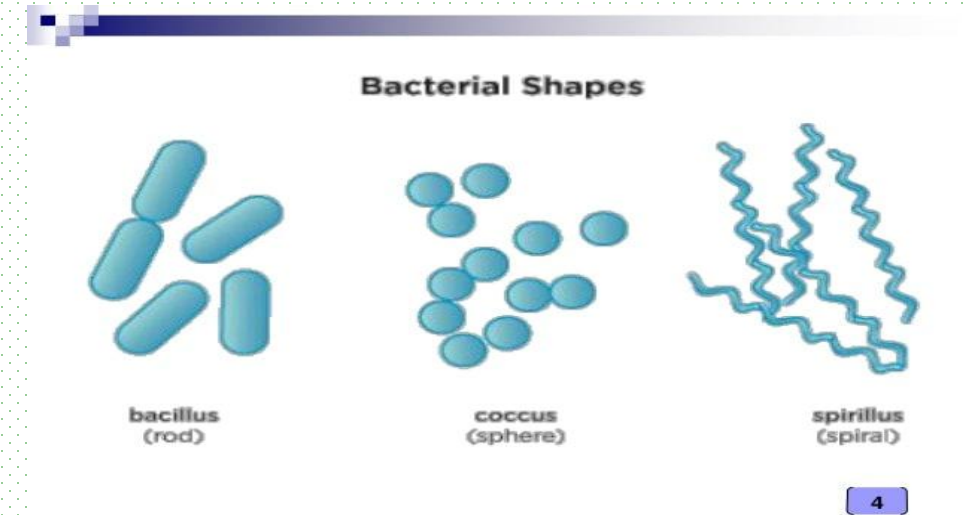
- All bacteria are **unicellular** organisms that reproduce by **binary fission**.
- Found in **diverse environments** (soil, water, living organisms)
- Bacterial cell are **extremely small** and are most conveniently measured in **micron** (10^6m). Bacteria range in size from about 0.2 to 5 μm
- Bacteria metabolize **aerobically and anaerobically** and can utilize a range of substrates.

Cell Morphology

Bacteria have characteristic shape. The common microscopic morphologies are:

- Spherical or ovoid (cocci)
- Rod shaped (bacilli)
- Spiral (Spirochete)

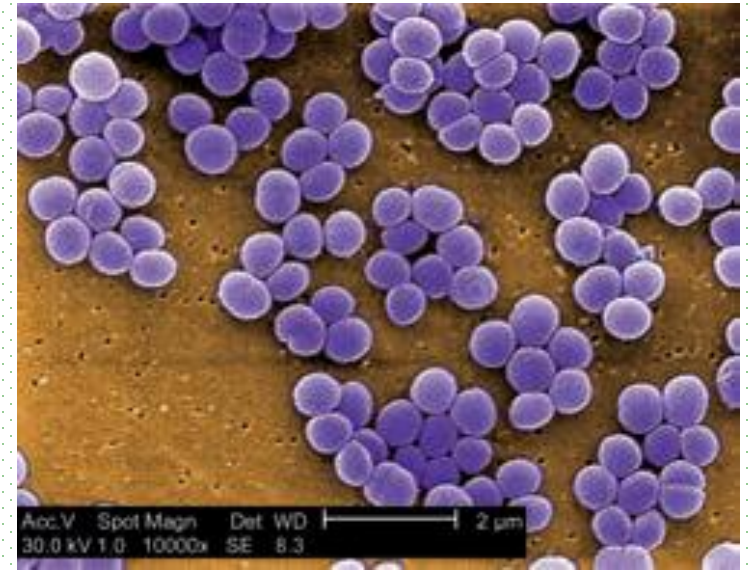
The microscopic appearance is valuable in classification and diagnosis



Arrangement of bacteria

- Pairs (diplococci)
- Chains (streptococci)
- Grapelike clusters (staphylococci).

These arrangements are determined by the orientation and degree of attachment of the bacteria at the time of cell division.



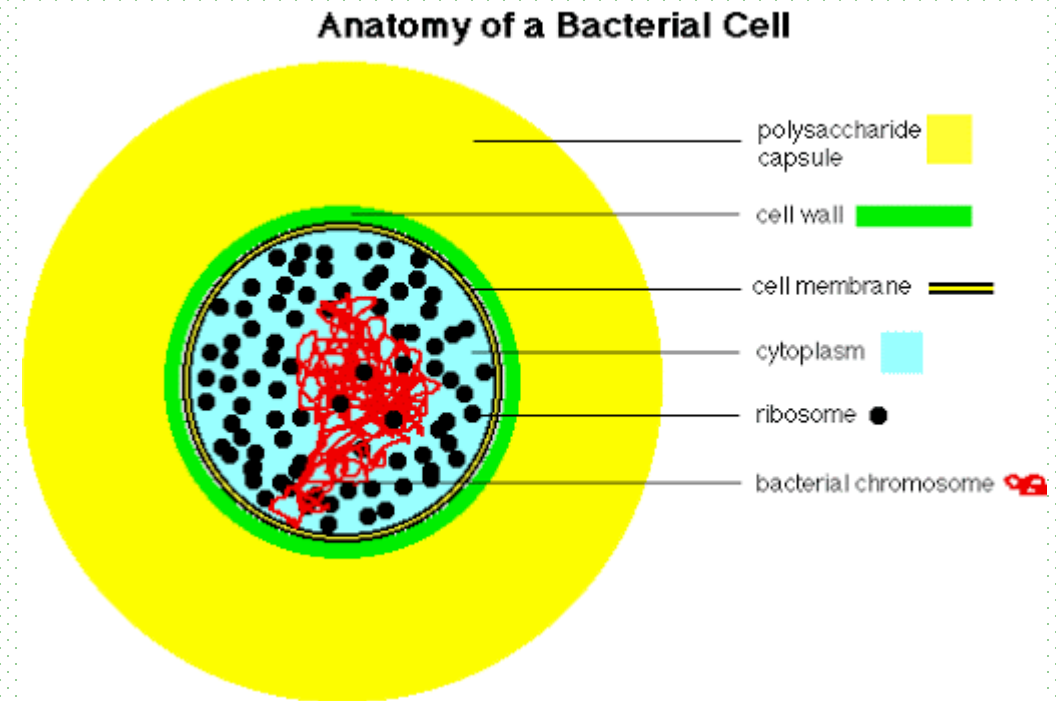
S. aureus—cocci in clusters



L. pneumophila—rods in chains

Structure of Bacterial cell

- Cell Wall
- Cytoplasmic Membrane
- Ribosome
- Nucleoid
- Structures Outside the Cell Wall
- Bacterial Spores



Structure of Bacteria

Cell Wall

- Outermost component common to all bacteria (except *Mycoplasma* species).
Located external to the cytoplasmic membrane
- Some bacteria have surface features external to the cell wall, such as a capsule, flagella, and pili
- Composed of peptidoglycan. The peptidoglycan provides structural support and maintains the characteristic shape of the cell.
- Peptidoglycan is found only in bacterial cells. It is a network that covers the entire bacterium and gives the organism its shape. It is composed of a sugar backbone (glycan) and peptide side chains (peptido).

Cell Walls of Gram-Positive and Gram-Negative Bacteria

- The structure, chemical composition, and thickness of the cell wall differ in gram-positive and gram-negative bacteria
- The peptidoglycan layer is much thicker in gram-positive than in gram-negative bacteria.
- Many gram-positive bacteria also have fibers of teichoic acid, whereas gram-negative bacteria do not have teichoic acids.
- In contrast, the gram-negative bacteria have a complex outer layer consisting of lipopolysaccharide, lipoprotein, and phospholipid.

Lipopolysaccharide (LPS)

This layer is unique to gram-negative bacteria, which is absent in gram-positives. It consists of three parts:

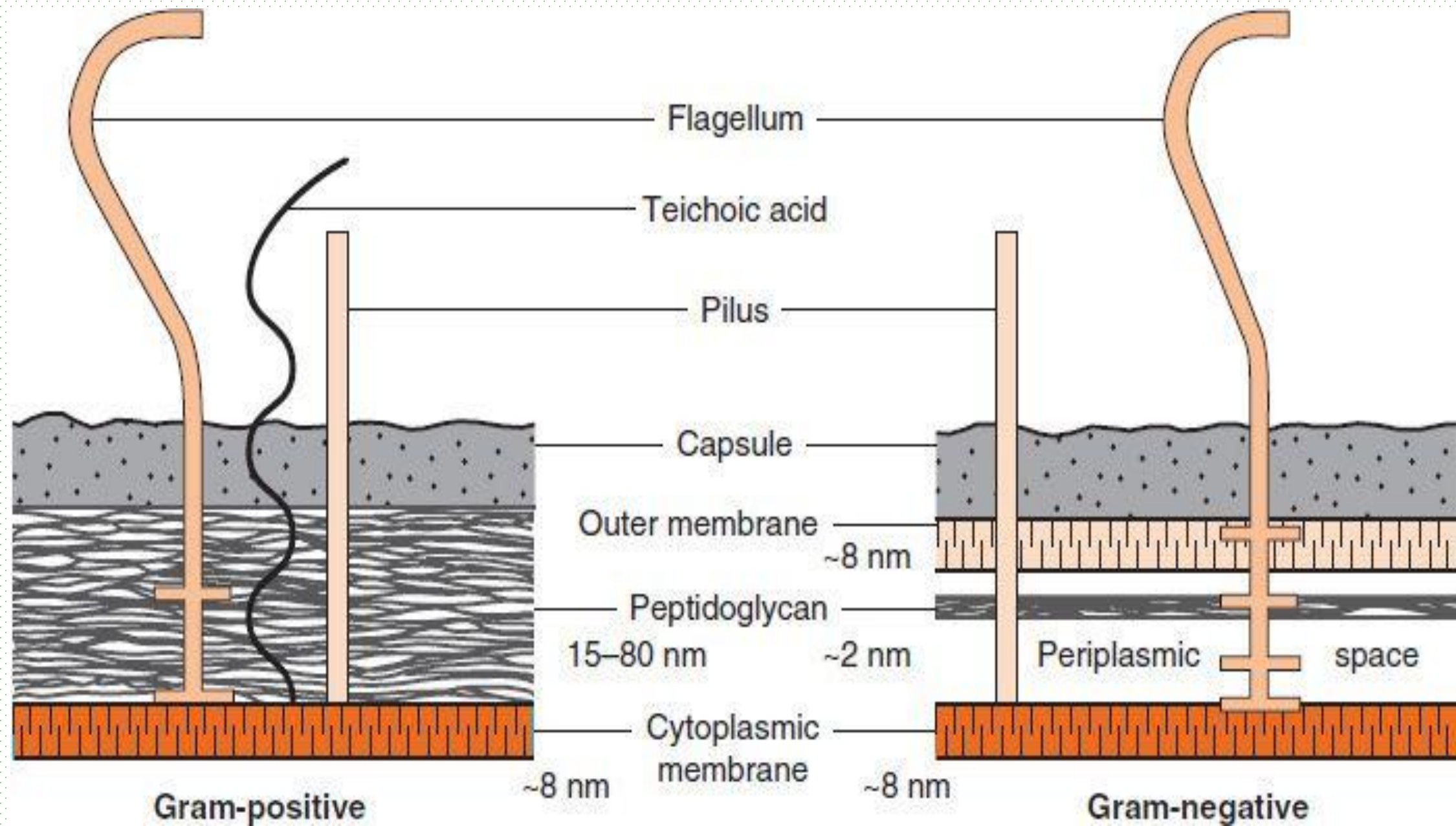
- ❑ **Lipid A or the endotoxin:** It has endotoxic activities, such as lethal effect, tissue necrosis, and antitumor activity
- ❑ **Core polysaccharide:** It is projected from lipid A region. It is composed of 10–12 sugar moieties
- ❑ **O side chain (or O antigen or somatic antigen):** It is a polysaccharide chain extending outwards from the core polysaccharide region. It greatly varies in composition between bacterial strains.

Comparison of Cell Walls of Gram-Positive and Gram-Negative Bacteria

Component	Gram-Positive Cells	Gram-Negative Cells
Peptidoglycan	Thicker; multilayer	Thinner; single layer
Teichoic acids	Yes	No
Lipopolysaccharide (endotoxin)	No	Yes

The cell wall has several other important properties:

- (1) The lipopolysaccharide of the outer membrane of the cell wall of gram-negative bacteria is endotoxin. It is responsible for many of the features of disease, such as fever and shock (is an integral part of the cell wall).
- (2) Its polysaccharides and proteins are antigens that are useful in laboratory identification.
- (3) Its porin proteins play a role in facilitating the passage of small, hydrophilic molecules into the cell. Porin proteins in the outer membrane of gram-negative bacteria act as a channel to allow the entry of essential substances as well as many antimicrobial drugs

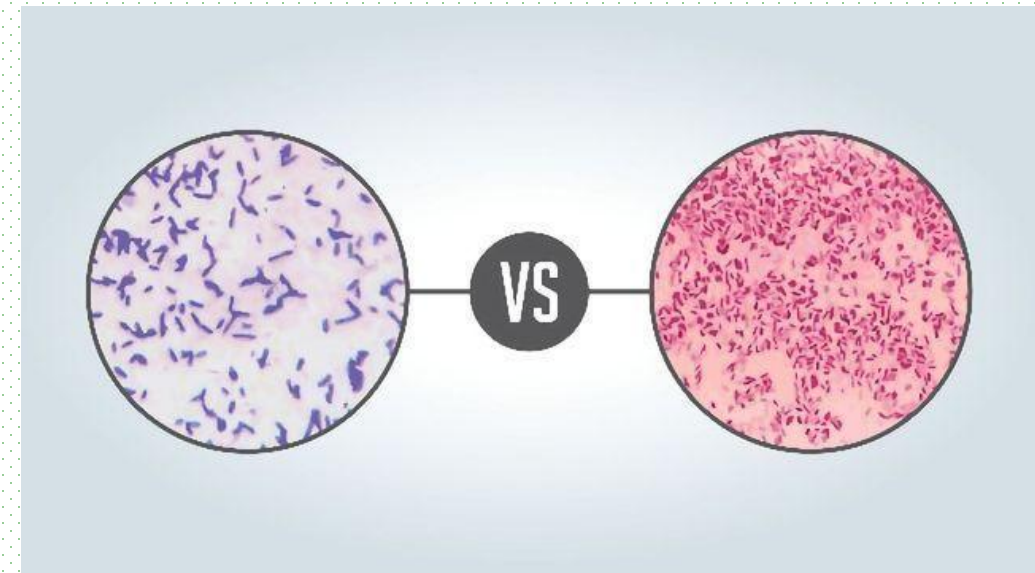


Gram stain

- Developed in 1884 by the Danish physician **Christian Gram**.
- Most important staining procedure in microbiology. It separates **most bacteria** into two groups: the **gram-positive bacteria**, which stain **purple**, and the **gram-negative bacteria**, which stain **red**.
- However, **not all bacteria** can be seen in the Gram stain.

The Gram stain is useful in two ways:

- (1) Identification of many bacteria.
- (2) Influencing the choice of antibiotic



Medically Important Bacteria That Cannot Be Seen in the Gram Stain

Name	Reason	Alternative Microscopic Approach
Mycobacteria, including <i>M. tuberculosis</i>	Too much lipid in cell wall so dye cannot penetrate	Acid-fast stain
<i>Treponema pallidum</i>	Too thin to see	Dark-field microscopy or fluorescent antibody
<i>Mycoplasma pneumoniae</i>	No cell wall; very small	None
<i>Legionella pneumophila</i>	Poor uptake of red counterstain	Prolong time of counterstain
Chlamydiae, including <i>C. trachomatis</i>	Intracellular; very small	Inclusion bodies in cytoplasm
Rickettsiae	Intracellular; very small	Giemsa or other tissue stains

Cell Walls of Acid-Fast Bacteria

- Mycobacteria (e.g., *Mycobacterium tuberculosis*) have an unusual cell wall, resulting in their inability to be Gram-stained.
- High concentration of lipids, called **mycolic acids**, in the cell wall of mycobacteria.

Note/ These bacteria are said to be **acid-fast** because they resist decolorization with acid–alcohol after being stained with carbolfuchsin.



Acid-fast stain of mycobacteria

Cytoplasmic membrane

Lying just beneath the cell wall, is a thin, flexible sheet surrounding the cytoplasm. Its structure is referred to as a fluid mosaic, in which the membrane is a continuous bilayer formed by phospholipids and proteins. The membrane has four important functions:

1. Active transport of molecules into the cell.
2. Energy generation by oxidative phosphorylation.
3. Synthesis of precursors of the cell wall.
4. Secretion of enzymes and toxins.

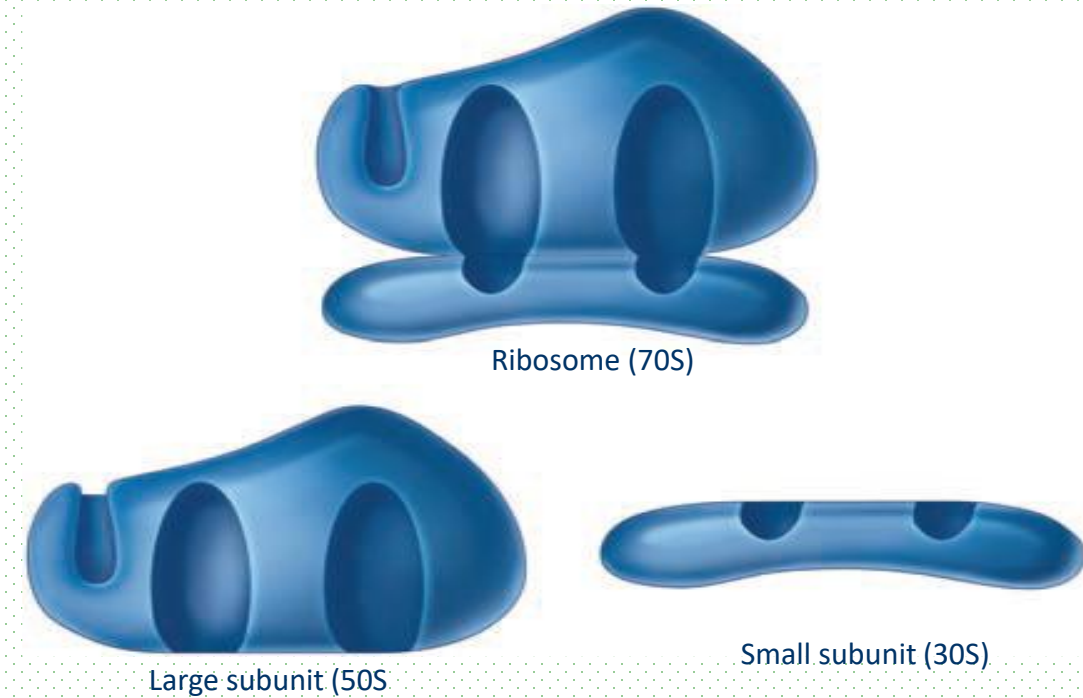
Cytoplasm

The cytoplasm has two distinct areas when seen in the electron microscope:

1. An amorphous matrix that contains ribosomes, nutrient granules, metabolites, and plasmids.
2. An inner, nucleoid region composed of DNA.

Ribosomes

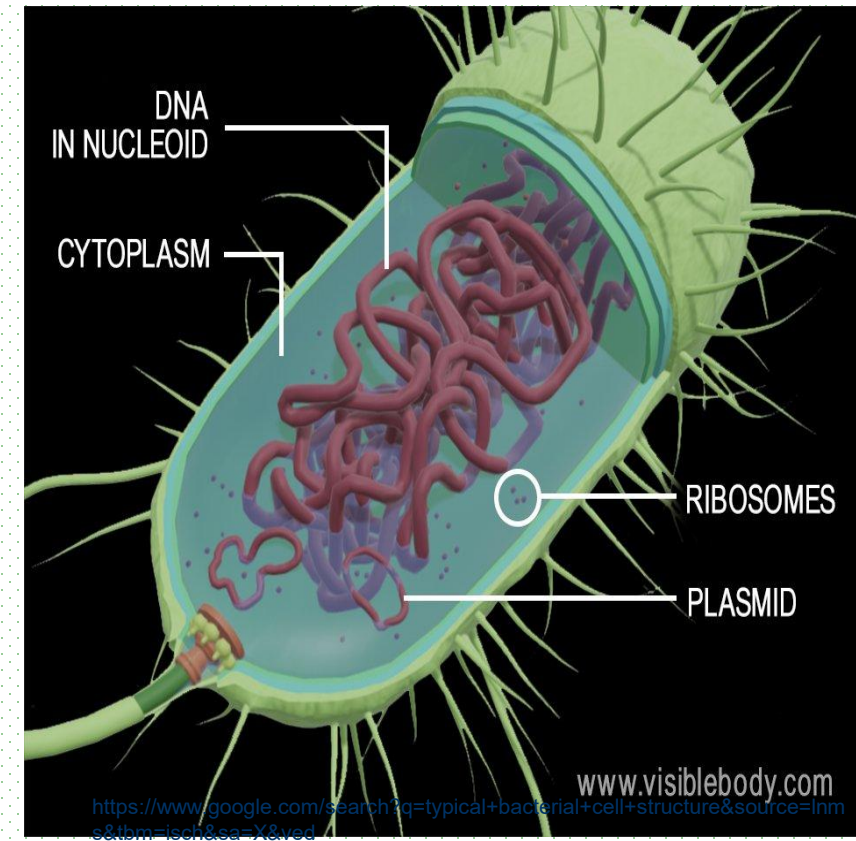
- Site of protein synthesis as in eukaryotic cells, but they differ from eukaryotic ribosomes in size and chemical composition.
- Bacterial ribosomes are 70S in size, with 50S and 30S subunits.



model of a prokaryotic ribosome, showing the small (30S) and large (50S) subunits, both separate and joined.

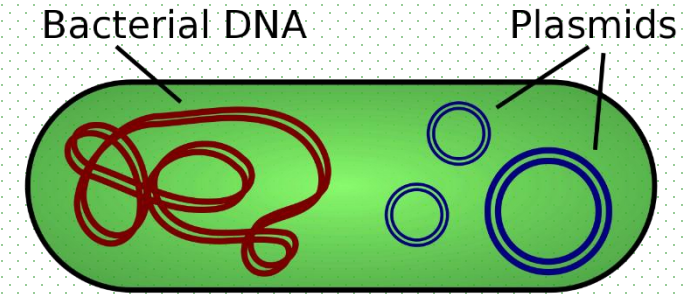
Nucleoid

- ❑ The genetic information of bacteria is carried in a **long, double-stranded (ds), circular molecule** of deoxyribonucleic acid (DNA).
- ❑ Bacteria do not have a **true nucleus**. Their DNA is not enclosed by a **nuclear membrane** but instead is aggregated in a central area of the cell called the nucleoid.
- ❑ The chromosome is an extremely long molecule of DNA that is tightly coiled to fit inside the cell compartment and divided into **genetic units (genes)** that carry information required for bacterial Maintenance and growth



Plasmids

- They are extrachromosomal, double-stranded, circular DNA molecules that are capable of replicating independently of the bacterial chromosome.
- Integrated into the bacterial chromosome.
- Occur in both gram-positive and gram-negative bacteria.
- Encode both exotoxins and many enzymes that cause antibiotic resistance



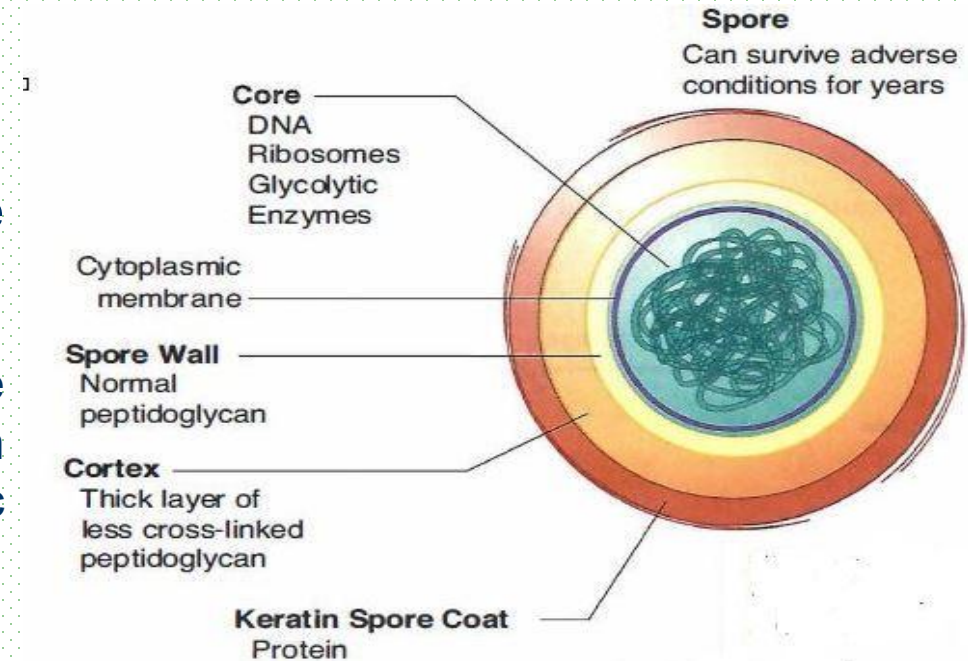
Transposons

- Small pieces of DNA that move frequently between chromosomal DNA and plasmid DNA. They carry antibiotic-resistant genes.
- Because of their unusual ability to move, they are nicknamed “jumping genes.”

Specialized Structures Outside the Bacterial Cell Wall

1- Capsule

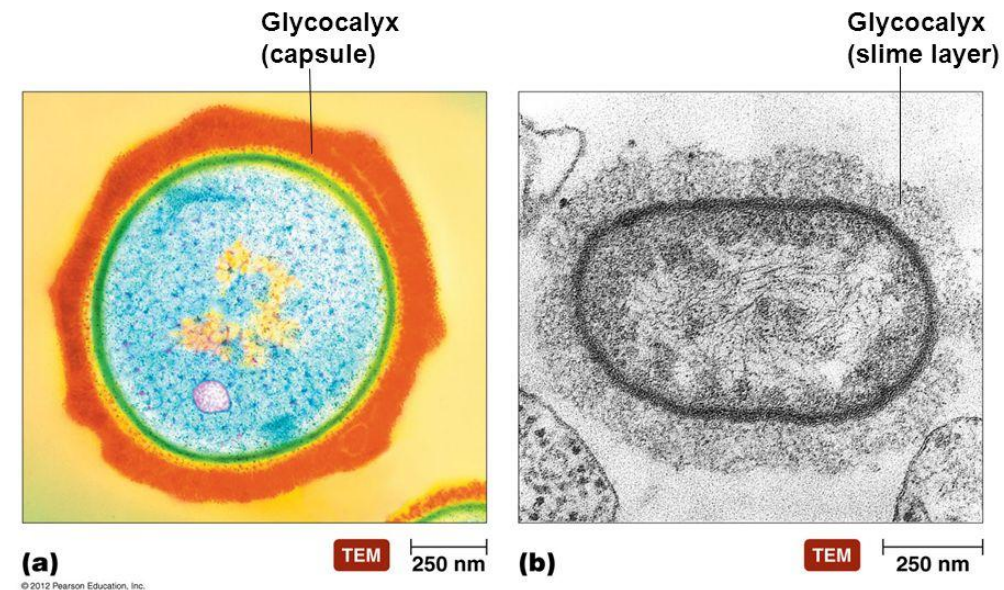
- Gelatinous layer covering the entire bacterium.
- Composed of polysaccharide, the polypeptide capsule of anthrax bacillus is the only exception (which has a capsule of polymerized d-glutamic acid).



2- Glycocalyx

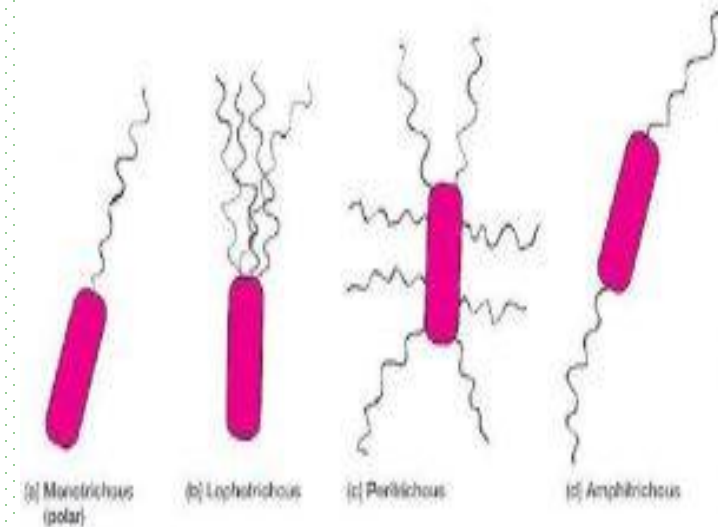
- Polysaccharide coating that is secreted by certain bacteria.
- Attaches bacteria firmly to the surface of human cells and to the surface of catheters, prosthetic heart valves, and prosthetic hip joints.

Figure 3.5 Glycocalyxes-overview



3- Flagella

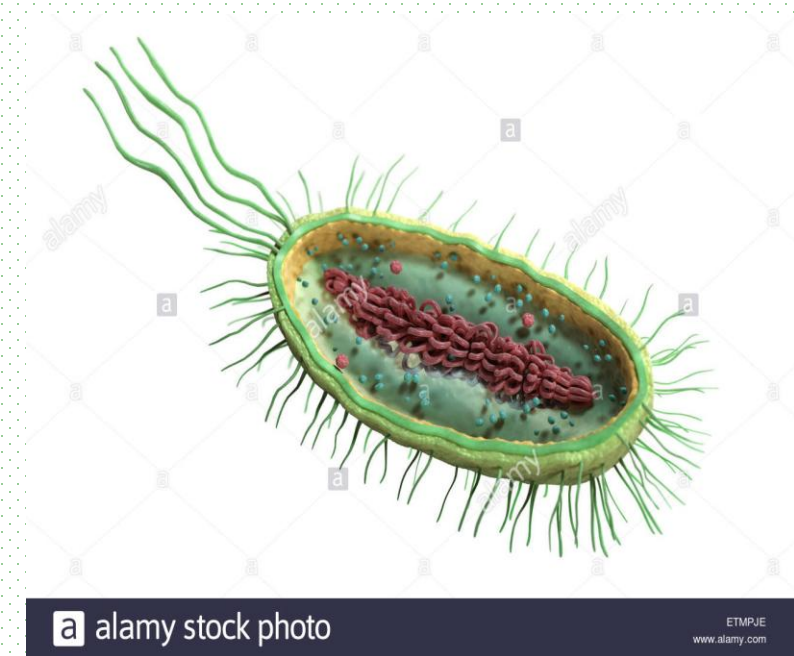
- Long, whiplike appendages that move the bacteria toward nutrients and other attractants, a process called chemotaxis.
- Consist of a basal body, hook, and a long filament composed of a polymerized protein called flagellin.
- Flagellated bacteria have a characteristic number and location of flagella. Flagella may be located in only one area of a cell (polar) or over the entire bacterial cell surface (peritrichous).



<http://microbeonline.com/bacterial-flagella-structure-importance-and-examples-of-flagellated-bacteria>

4- Pilli

- Hairlike filaments that extend from the cell surface.
- Shorter and straighter than flagella
- Composed of subunits of pilin, a protein arranged in helical strands.
- Found mainly on gram-negative organisms.



<http://www.alamy.com/stock-photo/nucleoid-dna.html>

Bacterial spores

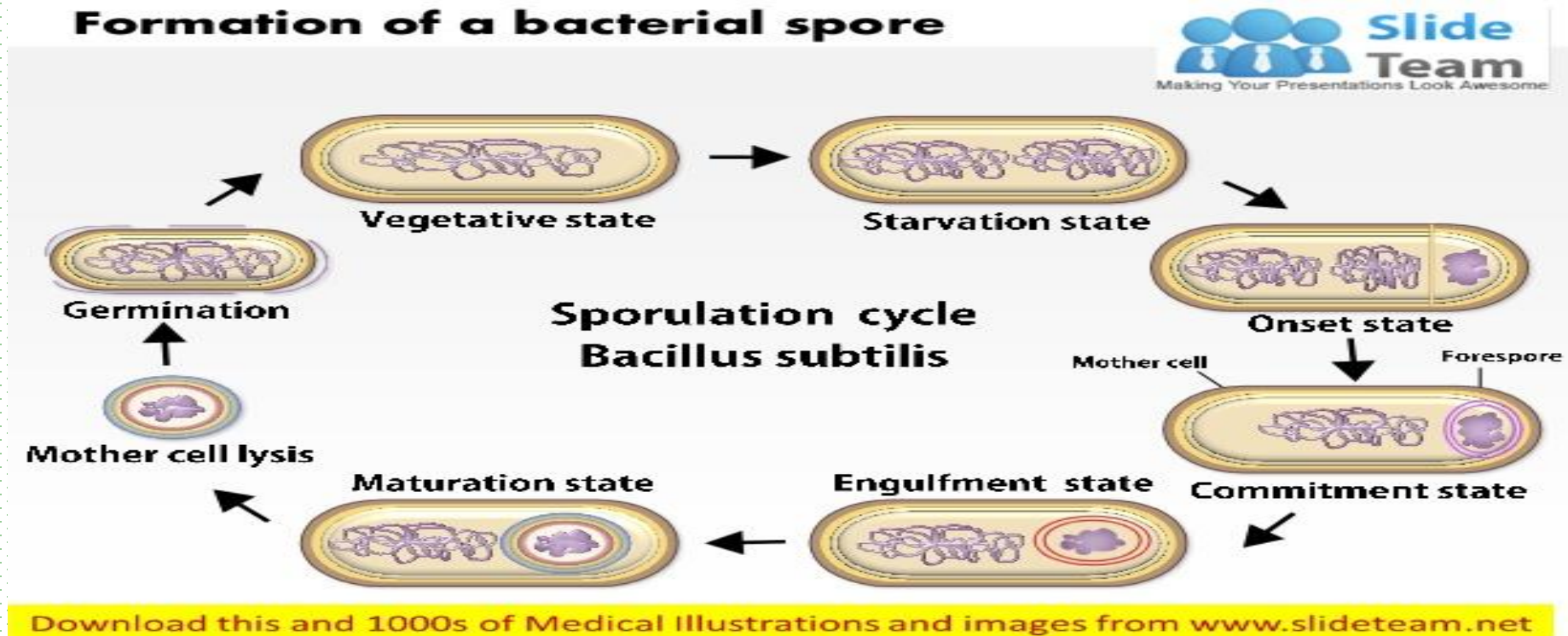
- These highly resistant structures are formed in response to adverse conditions.
- They are formed by certain gram-positive rods, especially *Bacillus* and *Clostridium* species.



Bacterial spores

- Sporulation occurs when nutrients, such as sources of carbon and nitrogen, are depleted.
- Each cell forms a single internal spore that is liberated when the mother cell undergoes autolysis.
- The spore is a resting cell, highly resistant to desiccation, heat, and chemical agents and can remain dormant for many years.
- Upon exposure to water and the appropriate nutrients, the spore germinates to produce a single vegetative cell.

Formation of bacterial spore



The medical importance of spores

- They are highly heat resistant and are not killed by many disinfectants. Boiling will not kill spores. Steam heating under pressure (autoclaving) at 121°C, usually for 30 minutes, is required to ensure the sterility of products for medical use.
- Spores are often not seen in clinical specimens recovered from patients infected by spore forming organisms.
- Sporulation is not a means of reproduction since one cell produces one spore that germinates into one cell.

MCQ

1/ Which reason explains why gram-positive bacteria are stain positive?

- A. Outer membrane absorbs the stain
- B. Acetone treatment precipitates intracellular stain.
- C. Heat fixing uncovers stainable proteins.
- D. Cell wall retains the crystal violet stain
- E. Bacteria reduce the stain to produce the characteristic color.

2/ Mycobacterium tuberculosis is called acid-fast because: -

- A. It is easily stained by gram stain
- B. Once stained, it can not be remove by acid
- C. It is easily decolorizes after staining.
- D. It is stained by an alkaline stain
- E. It is stained by an acid stain

3/ A 68-year-old woman on chemotherapy for leukemia has developed sepsis due to an infection with *Escherichia coli* (Gram negative bacteria). The following day the patient develops septic shock and dies. The structure on the bacterium most likely responsible for causing septic shock in this patient is

- (A) capsule
- (B) lipopolysaccharide
- (C) pili
- (D) spore
- (E) teichoic acid

4/ A 65-year-old man develops pneumonia. The organisms isolated from the sputum are gram-positive cocci that are alpha hemolytic on blood agar and sensitive to optochin. Which structure of the causal agent provides protection against phagocytosis?

- A. Capsule
- B. Pilli
- C. Coagulase
- D. Flagella
- E. Teichoic acid

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